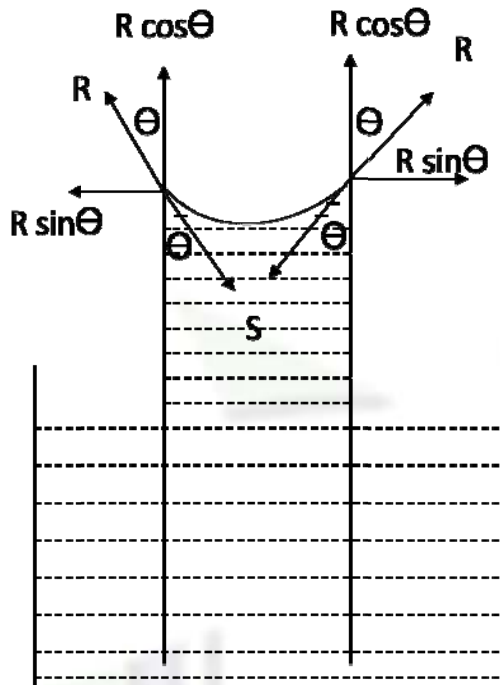


Surface Tension-Determination Of Surface Tension



Experimental determination of surface tension of water by capillary tube method



Given:

r = radius of the capillary tube

h = height of water level inside the capillary tube or height of water meniscus above the free surface of the liquid in the trough.

S = surface tension of the liquid

ρ = density of the liquid

Θ = angle of contact between the liquid and glass

At every point of contact in the free surface the force due to surface tension acts parallel to the free surface i.e. along the tangent as shown in the figure.

Since the force S exerted by the liquid molecules on the glass molecules at the point of contact hence due to reaction the glass molecules on the line of contact will also exert force R on the liquid molecules as equal and opposite reaction.

$$|\vec{R}| = |\vec{S}|$$

or $R = S$

If we resolve the force R into horizontal and vertical components the vertical component $R \cos \Theta$ at every points along vertically upward direction but the horizontal component $R \sin \Theta$ at diametrically oppositely directed. Hence when added over all the points on the boundary due to mutually cancellation in pairs the total horizontal force on the boundary is zero. The resultant force on the liquid surface is along vertically upward direction.

$$= S \cos \Theta \times \text{total length of the boundary}$$

$$= S \cos \Theta \times 2 \pi r$$

Due to this constant vertical upward force liquid rises in the capillary tube. As the liquid rises weight of the liquid comes into play along vertically downward direction and this force increases as the height increases. When the weight of the liquid becomes equal to the vertical force the resultant force becomes zero and the liquid does not rise any further.



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The weight of the liquid in the capillary tube = $\pi r^2 h \rho g$

From equation (1) and (2):

$$S \cos \theta \times 2 \pi r = \pi r^2 h \rho g$$

$$S = \frac{r h \rho g}{2 \cos \theta}$$

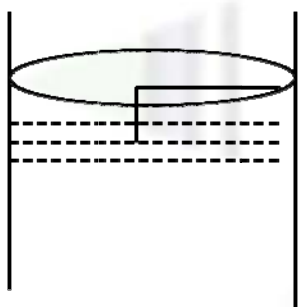
$$\left\{ \begin{array}{l} h = \frac{1}{r} \left(\frac{2S \cos \theta}{\rho g} \right) \\ h \propto \frac{1}{r} \end{array} \right.$$

Since for water in contact with clean glass θ very small

$$\cos \theta \approx \cos 0 \approx 1$$

$$S = \frac{r h \rho g}{2}$$

Correction: In determining the equation (4) we have neglected the weight of the liquid in the meniscus. Assuming that the radius of the meniscus is equal to the radius of the capillary tube.



Volume of the liquid in the meniscus

= Volume inside cylinder – Volume of the hemisphere

$$= \pi r^3 - \frac{1}{2} \frac{4}{3} \pi r^3 = \frac{\pi r^3}{3}$$

$$\text{Weight of the liquid in the meniscus} = \frac{\pi r^3 \rho g}{3}$$

$$\begin{aligned} \text{Total weight} &= \pi r^2 h \rho g + \frac{\pi r^3 \rho g}{3} \\ &= \pi r^2 \rho g \left[h + \frac{r}{3} \right] \end{aligned}$$

From equation (1) and (5):

$$2 \pi S \cos \theta = \pi r^2 \rho g \left[h + \frac{r}{3} \right]$$

$$S = \frac{r \rho g}{2 \cos \theta} \left(h + \frac{r}{3} \right)$$

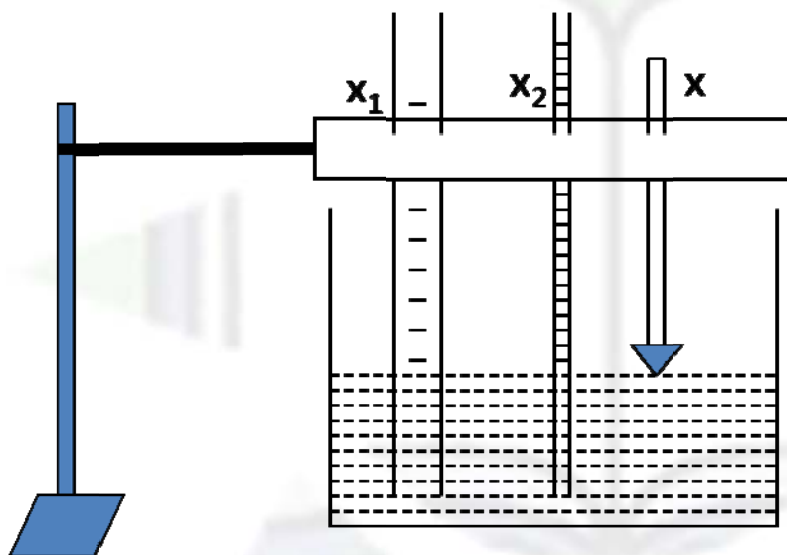
$$\text{Since } r \ll h \text{ hence } r/3 \text{ can be neglected } S = \frac{r \rho g h}{2 \cos \theta}$$

Surface Tension-Determination Of Surface Tension



Experimental arrangement:

- (1) Capillary tube of different radius is fixed on a glass plate. A needle of known length (h) is also fixed on the glass plate.
- (2) The glass plate is held on a stand so that capillary tube is exactly vertical.
- (3) The water is taken in a trough so that the capillary tubes dip in water and the lower tip of the needle just touches the surface of water in the trough.
- (4) The water meniscus in the capillary tubes is focused by a traveling microscope and the readings are noted.
- (5) The upper tip of the needle is focused by a traveling microscope and the reading is noted.
- (6) The radius of the capillary tubes is measured by the traveling microscope.



Calculation: Let x_1 , x_2 and x be the readings of meniscus when focused on the meniscus and upper tip of the needle respectively.

h = height of the needle

$$h_1 + h + (x_1 - x) = h_1 + x_1 - x$$

$$h_2 = h - (x - x_2) = h_1 + x_2 - x$$

Thus knowing h and r surface tension S can be calculated using equation (6)